

VAPO

Value-Model-Based Reinforcement Learning
for Advanced Reasoning

AIME 2024

60.4

60.4 on AIME 2024 — Surpassing DAPO by 10+ Points

ByteDance Seed • Base model: Qwen2.5-32B • No SFT data

arXiv: 2504.05118

Why Value Models? Finer Credit Assignment for Long CoT Reasoning

Value-Model-Free (GRPO / DAPO)

- Compute advantage from final trajectory reward
- Assign advantage uniformly across all tokens
- Simple — no value network to train
- Coarse credit assignment: every token gets the same signal
- Higher variance for long sequences
- Dominant approach (GRPO, DAPO, DeepSeek-R1-Zero)



Value-Model-Based (VAPO)

- Token-level credit assignment
- Each action's impact on returns is traced precisely
- Lower variance via learned baseline
- Generalizes across samples → better sample efficiency
- Critical when subtle step-level errors cause catastrophic failures
- Requires careful engineering (VAPO solves this)

Three Challenges Plaguing Value-Model-Based RL in Long-CoT

1

Value Model Bias

Long trajectories + bootstrapped learning cause compounding estimation errors in the value model.

2

Heterogeneous Sequence Lengths

Responses range from short to very long, each needing a different bias-variance tradeoff in advantage estimation.

3

Reward Signal Sparsity

Only the final <eos> token receives a reward signal — exacerbated by long chain-of-thought sequences.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Challenge 1 Value Model Bias

Value-Pretraining

Pre-train value model before RL to reduce initial estimation bias

Decoupled-GAE

Separate GAE computation for more robust value estimation

Challenge 2 Heterogeneous Lengths

Length-Adaptive GAE ★

Dynamic λ : low for short (less variance)
high for long (less bias)

Token-level Loss

Normalize loss per token, not per sequence (from DAPO)

Challenge 3 Reward Sparsity

Clip-Higher

Asymmetric clipping encourages exploration (from DAPO)

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling (from GRPO)

Sample multiple responses per prompt for better baseline estimation — benefits all three challenges

★ = novel contribution of VAPO

Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Novel VAPO contribution — addresses the heterogeneous sequence length problem unique to long-CoT reasoning

Short Responses → Lower λ

- Lower λ reduces variance in advantage estimation
- Short sequences have fewer tokens → Monte Carlo estimates are already reasonable
- Prioritize stability over bias correction

Long Responses → Higher λ

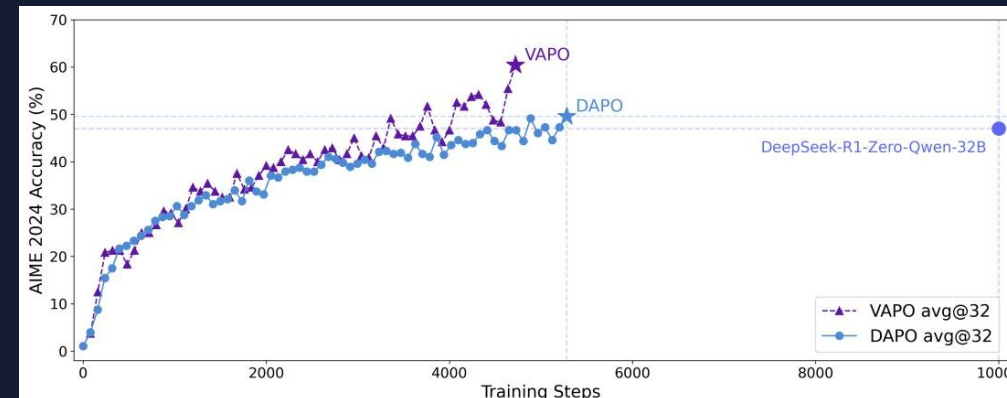
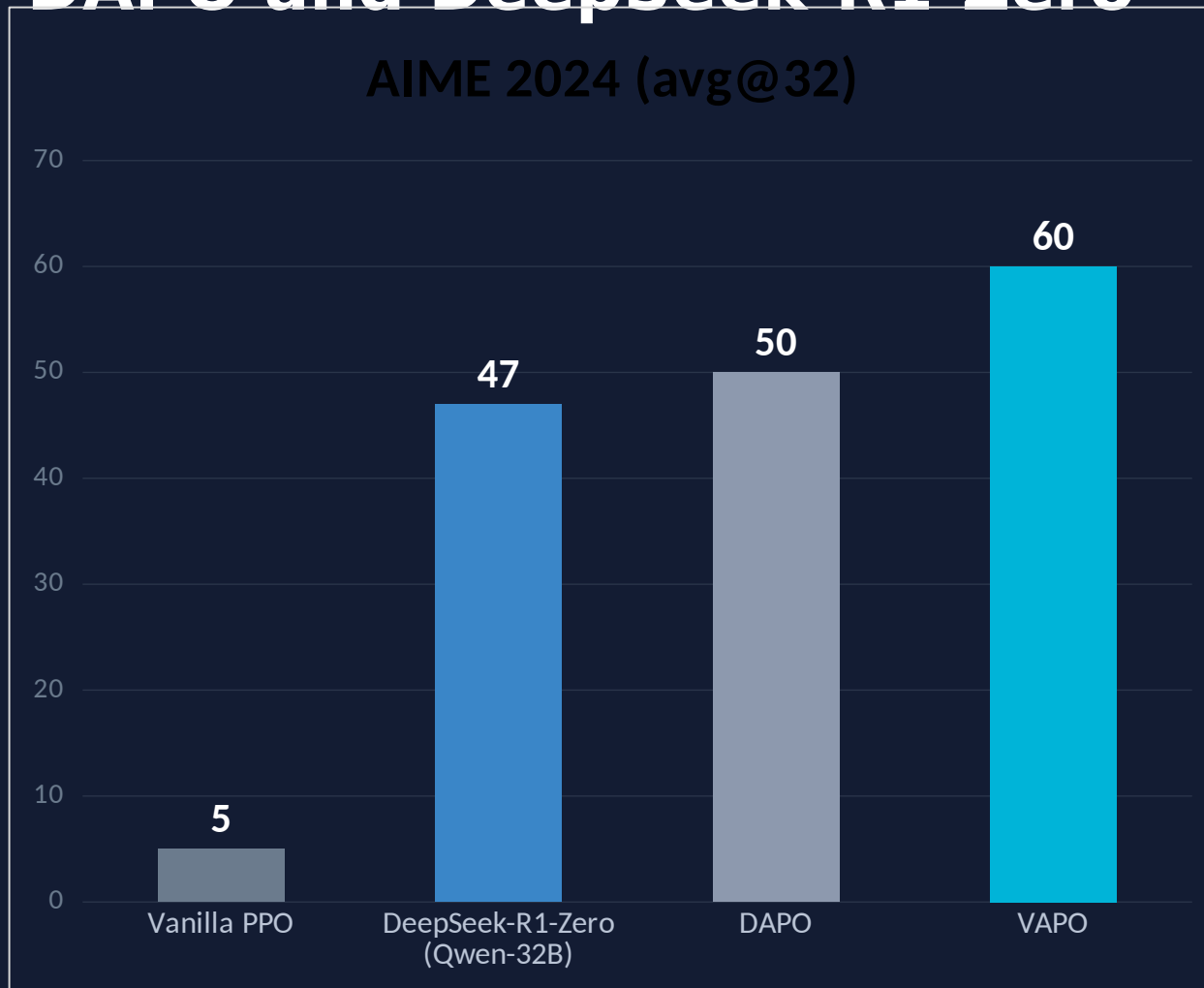
- Higher λ reduces bias from bootstrapped value estimates
- Long sequences accumulate compounding errors in value model
- Prioritize bias correction over variance reduction

Impact

Removing Length-Adaptive GAE drops AIME 2024 score from 60 → 45 (–15 points)

Optimizes bias-variance tradeoff per-sample, not one-size-fits-all — critical for heterogeneous CoT lengths

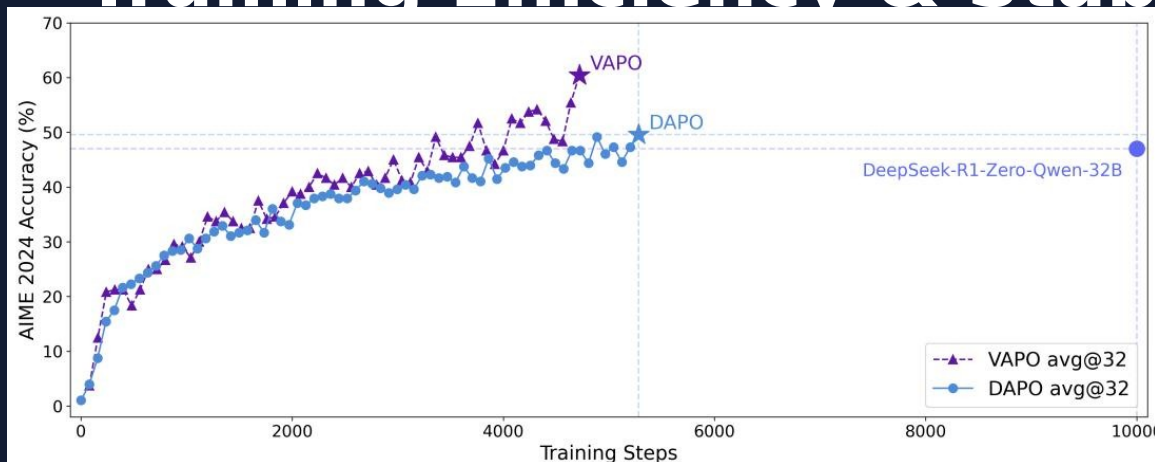
VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



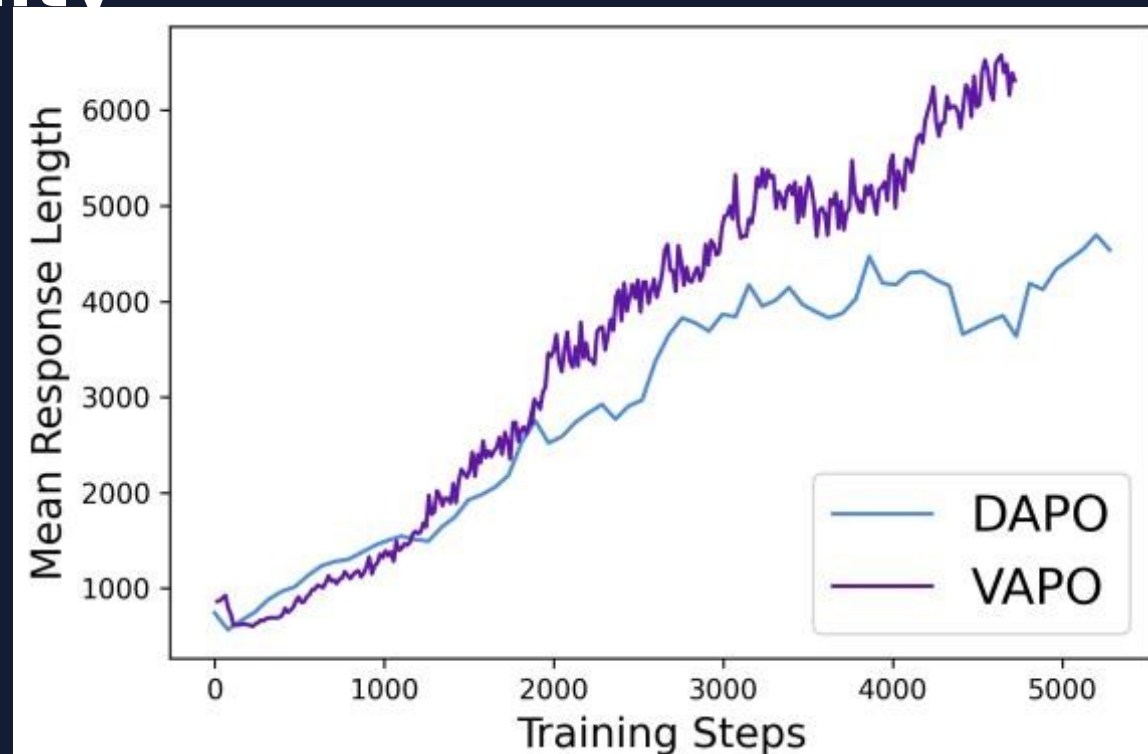
Training curve: VAPO vs DAPO on AIME 2024

All models trained on Qwen2.5-32B base, no SFT data • VAPO reaches 60% around step 4,500; DAPO plateaus near 50% at step 5,500

VAPO Reaches SOTA in ~5K Steps — Superior Training Efficiency & Stability



AIME 2024 Accuracy Over Training Steps



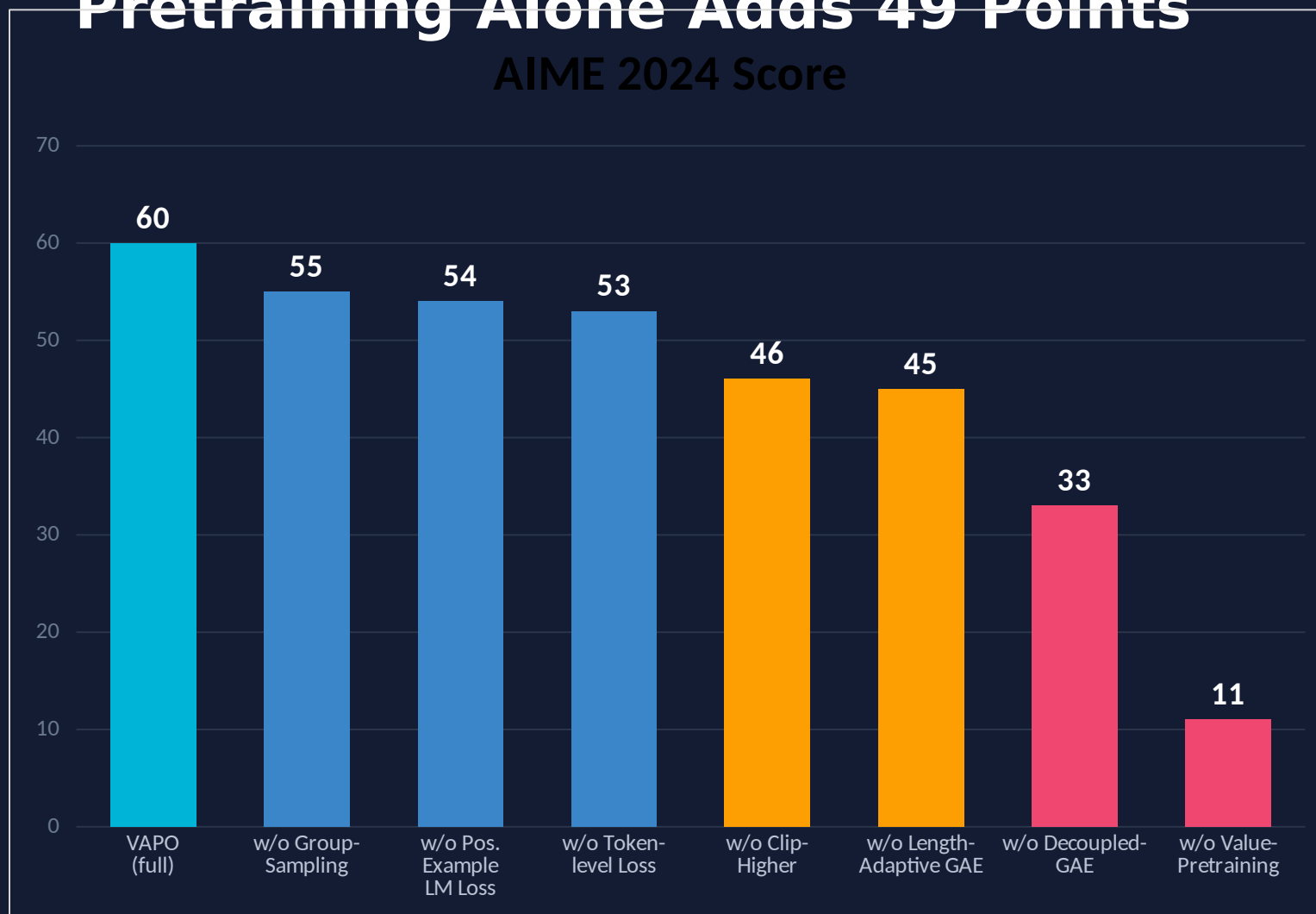
Mean Response Length Over Training

VAPO: 60% accuracy at ~4,500 steps • DAPO: plateaus at ~50% at step 5,500 • R1-Zero: 47% after ~10K steps

VAPO generates longer responses (~6,000 tokens vs ~4,500 for DAPO) — learns more elaborate reasoning chains • Zero training crashes across multiple runs

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

AIME 2024 Score



−49 pts

Value-Pretraining is the single most critical component (60 → 11)

−27 pts

Decoupled-GAE: robust value estimation is essential (60 → 33)

−15 pts

Length-Adaptive GAE: novel technique for heterogeneous lengths (60 → 45)

**−5 to
−7 pts**

Clip-Higher, Token-level Loss, Pos. Example LM Loss, Group-Sampling

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

1 Value-model-based RL can surpass value-free methods

VAPO (60.4) outperforms DAPO (50), GRPO-based DeepSeek-R1-Zero (47), and Vanilla PPO (5) — all on Qwen2.5-32B with no SFT data. The key: systematically address the three core challenges.

2 Value-Pretraining is the single most impactful component

Contributes 49 points (11 → 60). Without it, value-model-based RL barely works. This is the essential foundation for making value models viable in long-CoT settings.

3 Length-Adaptive GAE — a novel, critical technique

Dynamic λ per response length optimizes the bias-variance tradeoff for heterogeneous CoT sequences. Contributes 15 points and addresses a problem unique to long-form reasoning.

4 VAPO is training-efficient and remarkably stable

Reaches SOTA in ~5,000 steps with zero training crashes across multiple independent runs. DAPO requires ~5,500 steps for a lower score; R1-Zero needs ~10,000 steps.

5 Implications for future reasoning model development

Value-model-based RL is a viable — and potentially superior — paradigm for complex reasoning. The engineering framework (pretraining + adaptive GAE + decoupled estimation) is transferable.